

Bit-Plane B

DP8 (8×4) — decomposition over the bit planes of B

Step 1 — the dot product

$$\sum_{i=0}^7 A_i B_i$$

Step 2 — expand B into its bit planes and exchange the sums

Each B_i is a 4-bit value, $B_i = \sum_{j=0}^3 2^j B_i^j$, so

$$\sum_{i=0}^7 A_i B_i = \sum_{i=0}^7 \left[\sum_{j=0}^3 2^j B_i^j \right] A_i = \sum_{j=0}^3 2^j \sum_{i=0}^7 B_i^j A_i$$

$B_i^j A_i$ is a selection, not a product.

Step 3 — pair the lanes

$$\sum_{i=0}^7 A_i B_i = \sum_{j=0}^3 2^j \sum_{i=0}^3 \underbrace{\left[B_{2i}^j A_{2i} + B_{2i+1}^j A_{2i+1} \right]}_{R(B_{2i}^j, B_{2i+1}^j)}$$

$R(B_{2i}^j, B_{2i+1}^j)$ depends on the two bits B_{2i}^j, B_{2i+1}^j and takes one of four values:

B_{2i}^j	B_{2i+1}^j	$R(B_{2i}^j, B_{2i+1}^j)$
0	0	0
0	1	A_{2i+1}
1	0	A_{2i}
1	1	$A_{2i} + A_{2i+1}$

Step 4 — split the work

Every entry of the table is a function of A alone, so the table is built outside the PE and only the selection stays inside:

$$\underbrace{\{0, A_{2i}, A_{2i+1}, A_{2i} + A_{2i+1}\}}_{\text{dispatcher}} \longrightarrow \underbrace{\sum_{j=0}^3 2^j \sum_{i=0}^3 R(B_{2i}^j, B_{2i+1}^j)}_{\text{PE}}$$

B is the narrow operand: 4 planes × 4 lane pairs = 16 selections of 10 bits per DP8.